ARCHITECTURE AND DECISION RECORD

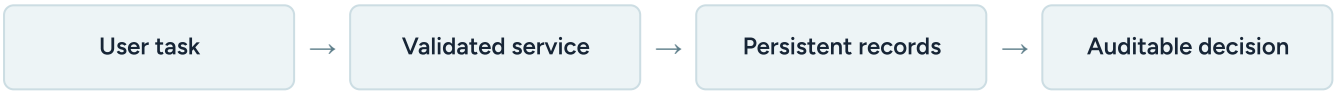
An accessible course resource discovery hub

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Accessibility | User research | React | Semantic HTML

Problem and contribution

This original design case study explores how to make approved course resources easier to discover and use while documenting accessibility support and unresolved limitations. I authored the requirements breakdown, architecture boundaries, data relationships, failure scenarios, and review rationale presented here. The design is a fictional presentation example, with no claim of a production deployment or measured client result.



Architecture decisions and tradeoffs

- Treat accessibility as an ongoing content and interaction responsibility rather than one pass/fail badge.
- Use semantic headings and native controls before adding custom keyboard behavior.
- Store barrier descriptions with restricted visibility and avoid collecting medical diagnoses.

Core records

Record	Responsibility
courses	A course contains many topics and reviewed resources.
resources	A resource has versions and linked accessibility metadata.
resource_versions	Each revision preserves provenance and supports review of changes.
accessibility_features	Features describe checks performed; they are not a universal accessibility guarantee.

The service layer owns authorization and business constraints. Interface validation improves the experience but cannot replace server checks. Event records preserve attributable decisions and their timing without logging unnecessary private content.

VERIFICATION AND REVIEW PRACTICE

Failure cases, evidence, and limitations

Proposed verification matrix

Scenario	Method	Expected behavior
Keyboard journey	Navigate course search, filters, results, save controls, and the report form without a mouse.	All actions are reachable with visible focus and no keyboard trap.
Reading order	Use a screen reader with headings and form labels to complete the same task.	The order, link purpose, status messages, and error recovery remain understandable.
Responsive reading	Enlarge text and test a narrow viewport without changing browser zoom assumptions.	Core content remains readable without lost controls or horizontal page scrolling.
Inclusive findability	Ask participants with varied access needs to locate a specific supported resource.	Record task success, barriers, and participant preferences without generalizing beyond the sample.

Technical review

Do not equate an automated accessibility score with successful student use. Evaluate keyboard and screen-reader journeys, document remaining barriers, and describe the recruitment limits of a small usability study.

How to evaluate the design

Report task success, accessibility issues by severity, keyboard steps, and time to locate a resource. Separate automated findings from human review. Describe recruitment and assistive technology settings so another reviewer can understand what was and was not evaluated.

The matrix specifies intended tests, not completed test results. A reviewer should capture the environment, initial records, inputs, expected state, observed state, and any unresolved defect before interpreting the outcome. Repeated runs should use separate unique request keys where relevant and preserve failures for analysis.

Scope limitations

No claim of legal accessibility certification, automated accommodation decisions, diagnosis collection, or redistribution of third-party copyrighted course packs.

Review the actual tasks and the content-review process; a high automated score does not by itself show that students can use the materials.

> **Authorship and reuse**
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> Prepared by Sofia Mendoza as an original presentation work sample. All narrative, schemas, and test scenarios are original illustrative material. There are no external client documents, employment certificates, real personal records, or fabricated citations in this case study.